BidVerify AI — 50-Requirement Stress-Test Tender

SYNTHETIC TEST DOCUMENT • designed to test messy-document requirement extraction and semantic matching

Tender: Integrated Solar Smart Streetlight & Remote Monitoring System

Test Tender ID: GEM-TEST-SSLM-2026-50

Bid window: 10/09/2026 to 20/09/2026

Quantity: 200 node-equivalents

Important: This is a synthetic benchmark created for BidVerify AI testing. It is not an actual government tender and does not represent an official GeM procurement document.

Messy-document characteristics intentionally included

The document contains headings, background prose, notes, duplicated concepts, mixed units, tables, optional/preferred clauses, exception-like wording, and requirements spread across sections. Only the 50 numbered T01–T50 items below are the ground-truth requirements.

Background note: The purchaser expects energy-efficient lighting and centralized monitoring. This paragraph is explanatory and is not itself a requirement.

Reference note: Terms such as node, controller, gateway and luminaire may be used interchangeably in supporting descriptions where context permits. This note is not a requirement.

Evaluation note: The buyer may request clarification where submitted evidence is ambiguous. This sentence is process guidance, not a technical specification.

Requirements 1–10

Code	Specification / Parameter	Required tender specification	Condition	Obligation
T01	System capacity	System shall support not less than 200 solar streetlight nodes in one managed deployment.	MINIMUM	MANDATORY
T02	LED luminaire output	Luminaire shall provide minimum 12,000 lumens at rated operating conditions.	MINIMUM	MANDATORY
T03	LED efficacy	LED efficacy shall be at least 150 lm/W.	MINIMUM	MANDATORY
T04	Color temperature	CCT shall be within 4000 K to 5000 K.	RANGE	MANDATORY
T05	Color rendering	CRI shall be minimum 70.	MINIMUM	MANDATORY
T06	LED lifetime	Rated LED lifetime shall be at least 50,000 hours.	MINIMUM	MANDATORY
T07	Ingress protection	Luminaire enclosure shall be minimum IP66.	MINIMUM	MANDATORY
T08	Impact protection	Luminaire impact resistance shall be minimum IK08.	MINIMUM	MANDATORY
T09	Solar panel capacity	PV array capacity shall be minimum 150 Wp per light.	MINIMUM	MANDATORY
T10	PV efficiency	Solar module efficiency shall be minimum 19%.	MINIMUM	MANDATORY

Supporting prose: The wording above should be interpreted together with applicable site conditions, drawings and purchaser instructions. Such explanatory text must not be extracted as an additional requirement.

Requirements 11–20

Code	Specification / Parameter	Required tender specification	Condition	Obligation
T11	Battery capacity	Battery bank shall provide minimum 100 Ah at the specified system voltage.	MINIMUM	MANDATORY
T12	Battery chemistry	Battery shall be LiFePO4 or an equivalent chemistry approved by the purchaser.	ALLOWED	MANDATORY
T13	Autonomy	System shall provide minimum 3 nights of autonomy without solar charging.	MINIMUM	MANDATORY
T14	Dimming	Controller shall support programmable 0-100% dimming.	RANGE	MANDATORY
T15	Dusk-to-dawn	Automatic dusk-to-dawn operation shall be supported.	QUALITATIVE	MANDATORY
T16	Motion sensing	Motion sensor shall support detection up to 10 m.	MINIMUM	MANDATORY
T17	Operating temperature	Outdoor electronics shall operate from -20°C to +55°C.	RANGE	MANDATORY
T18	Wind load	Pole/luminaire assembly shall withstand design wind speed of at least 150 km/h.	MINIMUM	MANDATORY
T19	Remote monitoring	Platform shall provide remote monitoring of lamp status, battery status and faults.	QUALITATIVE	MANDATORY
T20	Communication	Controller shall support 4G/LTE connectivity and store data during temporary network loss.	QUALITATIVE	MANDATORY

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Requirements 21–30

Code	Specification / Parameter	Required tender specification	Condition	Obligation
T21	Data retention	Monitoring platform shall retain operational telemetry for at least 24 months.	MINIMUM	MANDATORY
T22	Dashboard	Web dashboard shall provide live status, alarms and historical trends.	QUALITATIVE	MANDATORY
T23	API	System shall expose a documented REST API for telemetry and alarms.	QUALITATIVE	MANDATORY
T24	GPS	Each node shall support GPS or equivalent location tagging.	QUALITATIVE	MANDATORY
T25	Time synchronization	Controllers shall synchronize time using NTP or an equivalent mechanism.	QUALITATIVE	MANDATORY
T26	Encryption in transit	Management traffic shall use TLS 1.2 or higher.	MINIMUM	MANDATORY
T27	Encryption at rest	Sensitive stored credentials/configuration shall be encrypted at rest.	QUALITATIVE	MANDATORY
T28	Authentication	Administrative access shall support role-based access control and MFA.	QUALITATIVE	MANDATORY
T29	Audit logs	Administrative and configuration actions shall be audit logged for at least 12 months.	MINIMUM	MANDATORY
T30	Password policy	Local accounts shall enforce minimum 12-character passwords where local authentication is used.	MINIMUM	MANDATORY

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Requirements 31–40

Code	Specification / Parameter	Required tender specification	Condition	Obligation
T31	Firmware update	Controller firmware shall support secure remote OTA updates with rollback.	QUALITATIVE	MANDATORY
T32	Fail-safe lighting	A communications outage shall not switch lights off; local schedules shall continue operating.	QUALITATIVE	MANDATORY
T33	Surge protection	AC/DC interface shall include minimum 10 kV surge protection.	MINIMUM	MANDATORY
T34	Power factor	AC auxiliary supply power factor shall be minimum 0.95.	MINIMUM	MANDATORY
T35	Energy metering	Controller shall record cumulative energy/operating-hour data per node.	QUALITATIVE	MANDATORY
T36	Installation	Supplier shall complete installation and commissioning within 60 days of site handover.	MAXIMUM	MANDATORY
T37	Training	Supplier shall provide at least 2 administrator training sessions.	MINIMUM	MANDATORY
T38	Warranty	Complete system warranty shall be minimum 5 years.	MINIMUM	MANDATORY
T39	Support response	Critical faults shall receive a support response within 4 hours.	MAXIMUM	MANDATORY
T40	Replacement SLA	Failed controller/luminaire components shall be replaced within 7 calendar days after confirmed failure.	MAXIMUM	MANDATORY

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Requirements 41–50

Code	Specification / Parameter	Required tender specification	Condition	Obligation
T41	Spare parts	Supplier shall maintain critical spare parts availability for at least 5 years.	MINIMUM	MANDATORY
T42	Documentation	Supplier shall provide installation, operation and maintenance manuals in English	QUALITATIVE	MANDATORY
T43	Compliance	LED driver/luminaire shall comply with applicable BIS/Indian standards relevant to the offered equipment.	QUALITATIVE	MANDATORY
T44	Environmental compliance	Battery disposal/recycling shall follow applicable environmental and e-waste/battery management requirements.	QUALITATIVE	MANDATORY
T45	Water ingress warranty	Warranty shall cover water-ingress failures of the luminaire housing under normal installation conditions.	QUALITATIVE	MANDATORY
T46	Price ceiling	Total quoted price for the base scope shall not exceed INR 25,00,000.	MAXIMUM	PREFERRED
T47	Delivery	First site delivery shall begin within 30 days of purchase order.	MAXIMUM	PREFERRED
T48	Local content	Bidder should disclose local content percentage for the offered equipment.	QUALITATIVE	PREFERRED
T49	MSE status	Bidder should provide applicable MSE/Udyam registration details if claiming MSE preference.	QUALITATIVE	PREFERRED
T50	Extended warranty	An optional extended warranty of at least 2 additional years should be available.	MINIMUM	PREFERRED

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